

University of Engineering and Technology Lahore

Section Course Outline Report

Department: Electrical Engineering

Printed Date: August 01, 2026

Section Course Detail					
Semester		Fall 2026			
Department		Electrical Engineering			
Program		BS AI			
Subject Title		Machine Learning			
Pre-requisites:		None			
Subject Domain		Computing			
Contact		maheen.zulfqar@uet.edu.pk			
Measurable Student Learning Outcomes					
CLOs	Description	PLOs	Domain	Domain Level	Assessments
CLO1	Describe basic machine learning concepts, theories and applications.	23-PLO02	Cognitive	2. Understand	Mid Term, Quiz
CLO2	Apply supervised, unsupervised and reinforcement learning techniques to solve problems of moderate complexity.	23-PLO03	Cognitive	3. Apply	Project Quiz, Assignment, Mid Term, Final Term
CLO3	Develop a reasonable size project using suitable machine learning technique.	23-PLO04	Cognitive	6. Create	Practical Problems and Project
Class Timings:					
Credit Hours: 3 hrs.					
Contact Hours: 3 hrs.					
Grading Policy					
Quiz1 10% Quiz2 10% Project 10% Mid Term 30%, Final 40%					

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Section Content		
Week (Lec)	Topics	CLO's
Week1	Introduction to Machine Learning • Artificial Intelligence, Machine Learning and Deep Learning • Applications of Machine Learning • Types of Machine Learning (Supervised, Unsupervised, Semi-Supervised, Reinforcement) • Machine Learning Pipeline • Introduction to Python Libraries	CLO1
Week 2	Data Preprocessing and Feature Engineering • Data Collection and Cleaning • Handling Missing Values • Feature Scaling and Feature Selection • Training, Validation and Test Sets • Python Practical Exercises	CLO1
Week 3	Regression and Classification Statistics Concepts Recap Population and Sampling Model Evaluation Matrices and Mathematical Concepts	CLO1 CLO2 CLO3
Week 4	Regression Techniques • Linear Regression • Multiple Linear Regression • Regression Performance Metrics (MAE, MSE, RMSE, R^2) • Python Practical Exercises	CLO1 CLO2 CLO3
Week 5	Logistic Regression and Classification • Logistic Regression • Confusion Matrix • Accuracy, Precision, Recall and F1-Score • Python Practical Exercises	CLO1 CLO2 CLO3
Week 6	Decision Trees • Tree Structure • Entropy and Information Gain • Gini Index, Overfitting, Underfitting, Bias-Variance Trade-off • Python Practical Exercises	CLO1 CLO2

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		CLO3
Week7	• Tree Pruning Bayes' Theorem • Conditional Probability • Gaussian & Multinomial Naïve Bayes, Applications • Python Practical Exercises	CLO1 CLO2
Week 8	K-Nearest Neighbors (KNN) • Distance Measures, Choosing K, Advantages, Limitations, Applications • Python Practical Exercises Review and Project Discussion • Real-world Case Studies • Project Discussion • Project Activity: Improve Kaggle score	CLO1 CLO2 CLO3
	Mid Term	CLO1, CLO2
Week 9	Support Vector Machines (SVM) • Hyperplanes and Margins • Kernel Functions • Advantages and Limitations • Python Practical Exercises	CLO1 CLO2 CLO3
Week 10	Ensemble Learning • Bagging • Random Forest • Boosting (Introduction) • Cross Validation and Model Comparison • Python Practical Exercises	CLO1 CLO2 CLO3
Week 11	Introduction to Unsupervised Learning • Clustering Concepts • Similarity Measures • Distance Metrics • Applications of Clustering • Python Practical Exercises	CLO1 CLO2 CLO3

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Week 12	Clustering Algorithms • K-Means Clustering • Hierarchical Clustering • Cluster Evaluation • Python Practical Exercises	CLO1 CLO2 CLO3
Week 13	Dimensionality Reduction • Curse of Dimensionality • Principal Component Analysis (PCA) • Feature Extraction vs. Feature Selection • Python Practical Exercises	CLO1 CLO2 CLO3
Week 14	Introduction to Artificial Neural Networks • Biological vs. Artificial Neurons • Perceptron • Multilayer Perceptron • Backpropagation (Concept) • Python Practical Exercises	CLO1 CLO2 CLO3
week15	Introduction to Reinforcement Learning • Agent and Environment • States, Actions and Rewards • Reinforcement Learning Workflow • Python Practical Exercises Project Evaluation	CLO1 CLO2 CLO3
week16	Basic Reinforcement Learning Algorithms • Q-Learning (Concept) • Exploration vs. Exploitation • Applications of Reinforcement Learning	CLO1 CLO2
	Final Term	CLO1, CLO2

Reference Materials: (or use any other standard and latest books)

1. Machine Learning, Tom, M., McGraw Hill, 1997.
2. **Géron, A. (2023).** Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow, 3rd Edition. O'Reilly Media.
3. Machine Learning: A Probabilistic Perspective, Kevin P. Murphy, MIT Press, 2012

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